Stat 122 (Mathematical Statistics II)

Lesson 3.3. Pivotal Quantities and Confidence Intervals

Learning Objectives

At the end of this lesson, students are expected to

1. describe a pivotal quantity and explain its role in statistical inference;
2. construct confidence interval estimates of common population parameters using pivotal quantities; and
3. interpret confidence intervals.

Introduction

A point estimator $\hat{\theta}$ provides a *one-shot guess* of the value of an unknown parameter θ. In statistics, we often face the challenge of making statements about an entire population when we only have access to a sample. How can we quantify the uncertainty in our estimates? This is where confidence intervals come in. An interval estimator, or confidence interval, provides a range of values that is likely to contain θ. Behind every confidence interval lies a powerful tool: the pivotal quantity. A pivotal quantity is a clever transformation of our data and parameters that has a distribution independent of the unknowns we're trying to estimate. By harnessing these quantities, we can construct confidence intervals that are both mathematically rigorous and practically useful.

Pivotal Quantity

Definition

A **pivotal quantity** ($Q(X, \theta)$) is a function of the observed data and unknown parameters whose probability distribution does not depend on those unknown parameters.

REMARK:

Pivotal quantities or simply a **pivot** is essential for constructing confidence intervals and hypothesis tests, because they allow inference without requiring knowledge of the true parameter values.

Example 3.3.1

The quantity

$$Q = Z = \frac{\overline{Y} - \mu}{\frac{\sigma_0}{\sqrt{n}}}$$

is a pivotal quantity since $Z \sim N(0, 1)$.

Example 3.3.2

Suppose $Y_1, Y_2, \cdots, Y_n$ follows a $U(0, \theta)$ distribution. Show that $Q = \frac{Y_{(n)}}{\theta}$ is a pivotal quantity.

SOLUTION:

We will apply the transformation method to find the distribution of $Q = \frac{Y_{(n)}}{\theta}$. With $q = \frac{y_{(n)}}{\theta}$, the inverse transformation is given by

$$y_{(n)} = q\theta$$

and the Jacobian is

$$dy_{(n)}/dq = \theta.$$

Thus, the PDF of Q, for values $0 < q < 1$, is given by

$$\begin{aligned} f_Q(q) &= f_{Y_{(n)}}(q\theta)|\theta| \\ &= n\theta^{-n}(q\theta)^{n-1}\theta, \ \text{ since } \ f_{Y_{(n)}}(y) = n\theta^{-n}y^{n-1}, \ \text{ for } \ 0 < y < \theta \\ &= nq^{n-1}. \end{aligned}$$

This shows that the distribution of Q is free of the parameter θ, thus, Q is a pivotal quantity. Note that the PDF of Q resembles the PDF of $Beta(n, 1)$, that is, $Q \sim Beta(n, 1)$.

Confidence Interval

Definition

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an *iid* sample from a population distribution (probability model) described by $f_Y(y, \theta)$. Informally, a confidence interval is an interval of plausible values for a parameter θ. More specifically, if θ is our parameter of interest, then we call $(\hat{\theta}_L, \hat{\theta}_U)$ a $(1-\alpha) \times 100\%$ confidence interval for θ if

$$P(\hat{\theta}_L \leq \theta \leq \hat{\theta}_U) = 1 - \alpha$$

where $0 < \alpha < 1$. We call $1-\alpha$ the confidence level. In practice, we would like the confidence level $1-\alpha$ to be large (say 0.90, 0.95, 0.99).

REMARK:

The interval $(\hat{\theta}_L, \hat{\theta}_U)$ is a random interval since these are functions of random observations. In other words, different samples will give different intervals $(\hat{\theta}_L, \hat{\theta}_U)$.

Steps in constructing confidence intervals using pivotal quantities

1. Identify the pivot (**Q**)

For location parameters (like a mean), we often use a *subtractive* pivot $(\bar{X} - \mu)$. For scale parameters (like variance), we use a *ratio* pivot (S^2/σ^2).

2. Determine the sampling distribution of **Q**
3. Set the probability limits

For a desired confidence level $(1-\alpha)$, find two critical values, a and b, from the tables of your identified distribution such that

$$P(a \leq Q \leq b) = 1 - \alpha$$

4. Rearrange algebraically the inequality to isolate the parameter in the middle

This is the most critical step. You must rearrange the inequality $a \leq Q(X, \theta) \leq b$ to isolate the parameter θ in the middle.

<u>Example 3.3.3</u>

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an *iid* sample from $N(\mu, \sigma_0^2)$, where the mean μ is unknown and variance σ_0^2 is known. The unknown parameter is μ. From Example 3.3.1

$$Z = \frac{\overline{Y} - \mu}{\frac{\sigma_0}{\sqrt{n}}}$$

is a pivotal quantity and $Z \sim N(0,1)$.

Using the standard normal distribution we have

$$\begin{aligned}
1-\alpha &= P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) \\
&= P\left(-z_{\alpha/2} \leq \frac{\overline{Y}-\mu}{\frac{\sigma_0}{\sqrt{n}}} \leq z_{\alpha/2}\right) \\
&= P\left(-z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}} \leq \overline{Y}-\mu \leq z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right) \\
&= P\left(z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}} > \mu-\overline{Y} > -z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right) \\
&= P\left(z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}} + \overline{Y} > \mu > -z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}} + \overline{Y}\right) \\
&= P\left(-z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}} + \overline{Y} < \mu < z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}} + \overline{Y}\right) \\
&= P\left(\overline{Y} - z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}} < \mu < \overline{Y} + z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right)
\end{aligned}$$

If we let $\hat{\theta}_L = \overline{Y} - z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}}$ and $\hat{\theta}_U = \overline{Y} + z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}}$, then

$$1-\alpha = P(\hat{\theta}_L < \mu < \hat{\theta}_U)$$

These calculations show that

$$\overline{Y} \pm z_{\alpha/2}\left(\frac{\sigma_0}{\sqrt{n}}\right)$$

is a $100(1-\alpha)\%$ confidence interval for the population mean μ.

<u>Example 3.3.4</u>

Suppose that Y is a single observation from an exponential distribution with mean θ.

a. Use the method of moment-generating functions to show that $\frac{2Y}{\theta}$ is a pivotal quantity and has a χ^2 distribution with 2 degrees of freedom.

b. Use the pivotal quantity $\frac{2Y}{\theta}$ to derive a 90% confidence interval for θ.

SOLUTION:

a. Since $Y \sim Exp(\theta)$, then

$$m_Y(t) = (1 - \theta t)^{-1}.$$

Let $U = \frac{2Y}{\theta}$. Then

$$\begin{aligned}
m_U(t) &= E[e^{tU}] \\
&= E[e^{t(2Y/\theta)}] \\
&= E[e^{(2t/\theta)Y}] \\
&= m_Y(2t/\theta) \\
&= [1 - \theta(2t/\theta)]^{-1} \\
&= (1 - 2t)^{-1}
\end{aligned}$$

The resulting MGF resembles the MGF for the chi–square distribution with 2 degrees of freedom. Thus, U has this distribution, and since the distribution of U does not depend on θ, U is a pivotal quantity.

b. A 90% confidence interval for θ is obtained as follows: $\alpha = 0.10$.

$$\begin{aligned}
0.90 &= P(a \leq Q \leq b) \\
&= P\left(\chi^2_{\nu=2,1-0.05} \leq \frac{2Y}{\theta} \leq \chi^2_{\nu=2,0.05}\right), \text{ since } Q \sim \chi^2_{\nu=2} \\
&= P\left(0.1025866 \leq \frac{2Y}{\theta} \leq 5.991465\right) \\
&= P\left(\frac{2Y}{5.991465} \leq \theta \leq \frac{2Y}{0.1025866}\right)
\end{aligned}$$